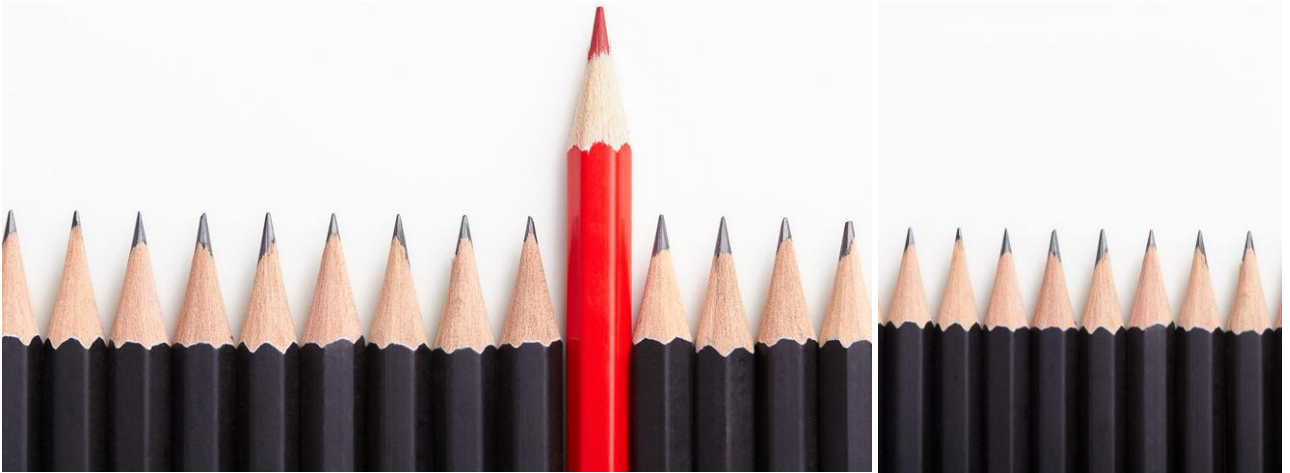




Identifying Assumptions and Market Types and MVP



1

Learning Outcomes



- Students should be able to
 - Explain the Lean startup approach
 - Explain the importance of phase 0 in the customer discovery process
 - State the following hypothesis.
 - Product Hypothesis
 - Customer Hypothesis
 - Distribution and pricing hypothesis
 - Market Type and hypothesis
 - Competitive Hypothesis
 - Explain the 4 Market types and why they matter
 - Explain build-Experiment-Learn Cycle
 - Explain the use of Minimum Viable Products
 - Explain Innovation Accounting

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Lean Startup - Value vs. Waste (Eric Ries)



Question: Which of our efforts are value-creating and which are wasteful?

Lean thinking defines value-creation as providing benefits to the customers; anything else is a waste!

But, how can you know whether you are providing benefits (i.e., creating value) to your potential customers?

Note: True startup productivity CANNOT be measured in terms of how much you are building every day, but rather in terms of systematically figuring out the right thing to build every day

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Lean Startup - Value vs. Waste (Eric Ries)



- Eric Ries's *Lean start up* has the goal to improve the success rate of new innovative products worldwide.
 - Eric Ries says the following about failure, “What makes these failures particularly painful is not just the economic damage done to individual employees, companies, and investors; they are also a colossal waste of our civilization’s most precious resource: the time, passion, and skill of its people”.
- Success is not about delivering a product; success is about delivering a product (or a feature of a product) that customers will use
 - The way to do this is to continuously align your efforts with your customers’ *real* needs
 - Note:** This is not about asking your customers what they need because customers typically do not know what they need

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Validated learning



- The most vital function of an IDE is learning.
 - Unfortunately, “learning” is the oldest excuse in the book for a failure of execution.
- Validated learning is not a sorry excuse, it is the intentional approach.
 - Validated learning is the process of demonstrating empirically that a team has discovered valuable truths about a startup’s present and future business prospects.
 - Involves having goals, e.g. What should we build and for whom? What market could we enter and dominate? How do we build sustainable value that will be eroded?

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Validated learning



- Creators fall in love with their products. They put in so many hours to build the perfect product that will change the world, then the launch the product and they sit down and wait for the customers to come in. A couple of days/weeks/month...the inflow of customers is not as expected.
 - They put in more hours to make a better product. A couple of days/weeks/month...the inflow of customers is not as expected.
- Waste is anything that does not create value for the customer.
 - While in traditional business, its easy to figure what the customer wants, But for the IDE, who the customer is and what the customer might find valuable are unknown,

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The Lean Canvas

Designed for: Designed by: Date: Version:

Problem	Solution	Unique Value Prop.	Unfair Advantage	Customer Segments
You complete the LC of BMC when seated in your office, Now its time to go to the field to test the assumptions gathered from secondary research, discussions and observations.				
Assumption Existing Alternatives	Assumption Key Metrics	Assumption High-Level Concept	Assumption Channels	Assumption Early Adopters
Assumption	Assumption	Assumption	Assumption	Assumption
Cost Structure		Revenue Streams		
Assumption		Assumption		

Lean Canvas is adapted from The Business Model Canvas (www.businessmodelgeneration.com/canvas). PowerPoint Implementation by: Neos Chronos Limited (<https://neoschronos.com/>). License: CC BY-SA 3.0

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Customer Discovery

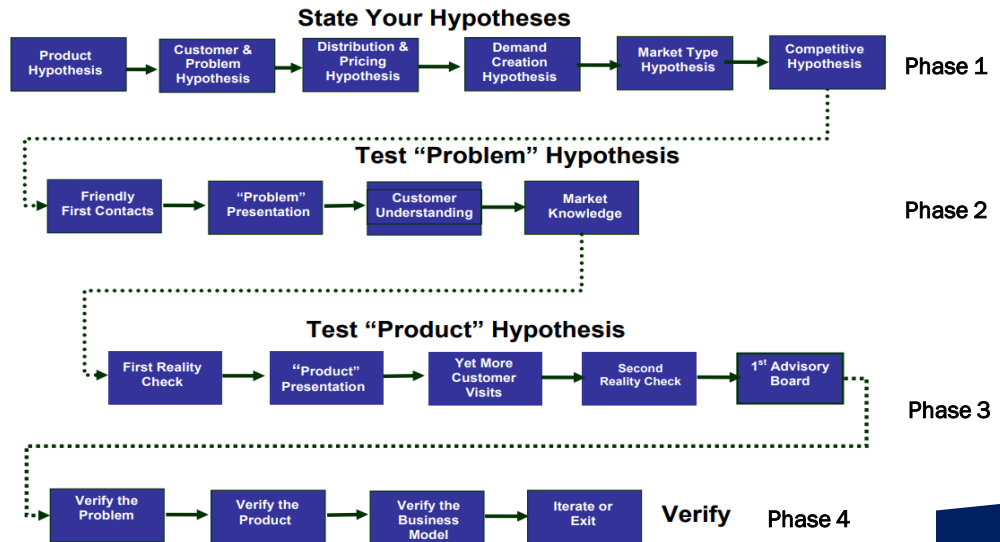


- The goal of Customer Discovery is just what the name implies: finding out who the customers for your product are and whether the problem you believe you are solving is important to them.
 - **This step involves discovering whether the problem, product and customer hypotheses in your business plan are correct.**
 - You need to leave guesswork behind and get “outside the building” in order to learn what the high-value customer problems are, what it is about your product that solves these problems, and who specifically are your customer and user (for example, who has the power to make or influence the buying decision and who actually will end up using the product on a daily basis.)
 - **The job of the Customer Development team is to see whether there are customers and a market for the product/service that you envision.**

Phases of the Customer Discovery process



Customer discovery is all about questioning your business's core assumption



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Phase o: Get Buy-In



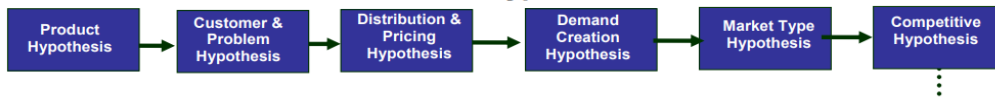
- Phase o consists of getting buy-in from all the key players.
- Customer Development as a separate process from Product Development is a new concept. Not everyone understands it.
- The Product Development process emphasizes execution. The Customer Development process emphasizes learning, discovery, and failure. For this reason you want to ensure there is enough funding for two to three passes through the Customer Discovery and Customer Validation steps.
 - Does the board believe Customer Development is iterative? Does the board believe it is necessary and worth spending time on?
 - Unique to the process is the commitment of the Product Development team to spend at least 15 percent of its time outside the building talking to customers.
- Create a mission statement- this may change over time.
- Write out core values- this don't change.

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Phase 1: State Your Hypotheses



State Your Hypotheses



- Hypotheses are assumptions that you have. Often these assumptions are educated guesses based on desktop research.
- Your written summary of these hypotheses will take the form of a one- or two-page brief about each of the following areas:
 1. Product
 2. Customer and their problem
 3. Channel and pricing
 4. Demand creation
 5. Market Type
 6. Competition

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THE FOUR TYPES OF STARTUP MARKETS



- There are four types of startups, and each of them has a very different set of requirements to succeed:
 - Startups that are entering an existing market
 - Startups that are creating an entirely new market
 - Startups that want to resegment an existing market as a low cost entrant
 - Startups that want to resegment an existing market as a niche player
- (see Clayton Christensen on Disruptive and Sustaining Innovations)

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THE FOUR TYPES OF STARTUP MARKETS



Customers	Market	Sales	Finance
Needs	Market Size	Distribution channel	On going capital
Adoption Rate	Cost of entry	Margins	Time to profitability
Problem recognition	Launch Type	Sales cycle	
Positioning	Competitive Barriers		

Table 2.1 Market Type Affects Everything

Market Type changes how you evaluate customer needs, customer adoption rate, how the customer understands his needs and how you would position the product to the customer. Market Type also changes the market size, as well as how you launch the product into the market.

Before any sales or marketing activities can begin, a company must keep testing and asking, "What kind of a startup are we?"

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THE FOUR TYPES OF STARTUP MARKETS



1. A New Product in an Existing Market

- An existing market is pretty easy to understand. We say you are in an existing market if your product offers higher performance than what is currently offered. Higher performance can be a product or service that runs faster, does something better or substantially improves on what is already on the market.
- The good news is that the users and the market are known, but so are the competitors. In fact, the competitors define the market. The basis of competition is therefore all about the product and product features.
- You can enter an existing market with a cheaper or repositioned "niche" product, but if that is the case we call that a resegmented market.

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THE FOUR TYPES OF STARTUP MARKETS



2. A New Product in a New Market (uber)

- What's a new market? It's what happens when a company creates a large customer base who couldn't do something before because of true innovation creating something never existed before, or dramatically lower cost that creates a new class of users. Or the new product solves availability, skill, convenience, or location issues in a way no other product has.
- In a new market the good news is that your product features are at first irrelevant because there are no competitors (except other pesky startups). The bad news is that the users and the market are undefined and unknown. If you're creating a new market, your problem isn't how to compete with other companies on product features but how to convince a set of customers that your vision is not a hallucination.
- Creating a new market requires understanding whether there is a large customer base who couldn't do this before, whether these customers can be convinced that they want or need your new product, and whether customer adoption occurs in your lifetime.
- It also requires rather sophisticated thinking about financing – how you manage the cash burn rate during the adoption phase, and how you manage and find investors who are patient and have deep pockets

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THE FOUR TYPES OF STARTUP MARKETS



3. A New Product Attempting to Resegment an Existing Market: Low Cost

- Resegmenting an existing market can take two forms: a low-cost strategy or a niche strategy.
- Low-cost resegmentation is just what it sounds like – are there customers at the low-end of an existing marketing who will buy “good enough” performance if they could get it at a substantially lower price? If you truly can be a low cost (and profitable) provider, entering existing markets at this end is fun, as incumbent companies tend to abandon low-margin businesses and head up-market.
- E.g Equity Bank
- While resegmenting an existing market is the most common Market Type choice of new startups, it's also the trickiest. As a low-end resegmentation strategy, it needs a long-term product plan that uses low cost as market entry to eventual profitability and up-market growth.

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THE FOUR TYPES OF STARTUP MARKETS



4. A New Product Attempting to Resegment an Existing Market: Niche

- Niche resegmentation is slightly different. It looks at an existing market and asks, “Would some part of this market buy a new product designed to address their specific needs? Even if it cost more? Or worse performance in an aspect of the product irrelevant to this niche. Niche resegmentation attempts to convince customers that some characteristic of the new product is radical enough to change the rules and shape of an existing market.
- Unlike low-cost resegmentation, niche goes after the core of an existing market’s profitable business.
- E.g In and Out Burger
- While resegmenting an existing market is the most common Market Type choice of new startups, it’s also the trickiest. As a niche resegmentation, this strategy faces entrenched competitors who will fiercely defend their profitable markets

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Market Type and the Customer Development Process



	Existing Market	Resegmented Markets	New Market
Customers	Existing	Existing	New/New usage
Customer Needs	Performance	1. Cost 2. Perceived need	Simplicity & convenience
Performance	Better/faster	1. Good enough at the low end 2. Good enough for new niche	Low in “traditional attributes”, improved by new customer metrics
Competition	Existing Incumbents	Existing incumbents	Non-consumption /other startups
Risks	Existing Incumbents	1. Existing incumbents 2. Niche strategy fails	Market adoption

Table 2.2 Market Type Characteristics

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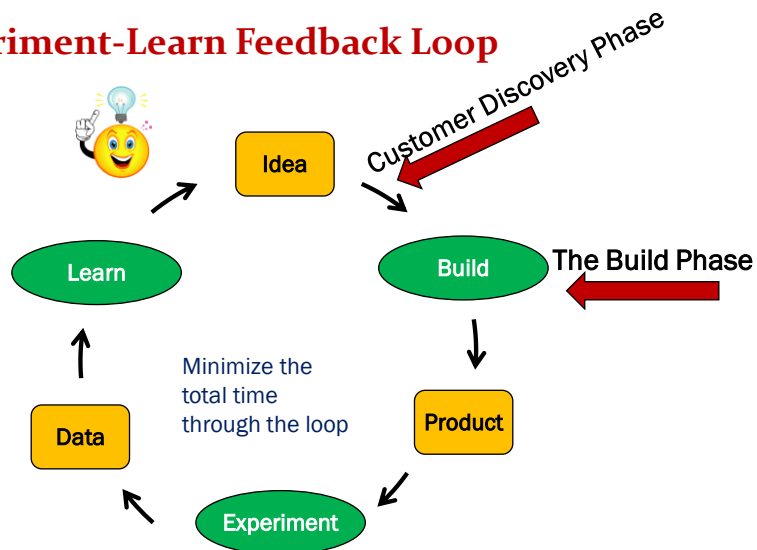


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Build-Experiment-Learn Feedback Loop



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The Build Phase: MVP



The build phase can be entered as quickly as possible with a **Minimum Viable Product** (MVP)

- An MVP ranges in complexity from extremely simple *smoke tests* (little more than an advertisement) to early prototypes and MVPs



- A **prototype** is a preliminary, interactive simulation used to test and refine design and functionality, focusing on user interaction and flow. An **MVP (Minimum Viable Product)** is a fully functional, real-world product with the core features needed to solve a user's problem, used to validate market demand and gather feedback from actual customers. **Prototypes are for internal design validation, while MVPs are for market validation.**

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The Build Phase: MVP



- Examples of MVP- Video MVP
 - Dropbox used a video to show what its product would look like and get customers to sign up to be notified when it was ready!
 - <https://www.youtube.com/watch?v=dR7tJ8wAl3M>
- THE CONCIERGE MINIMUM VIABLE PRODUCT
 - Replace the system with a person. A company that used this approach was Texas based startup Food on the Table. They build a mobile app that provides easy-to-cook recipes, sale items and grocery lists for weeknight diners.
 - Instead of building the app, they had someone do the whole process that they intended to automate, that person was the CEO.
 - They focused on one customer, then 2, then 3...

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Speed Bumps in Experimenting with an MVP



- Deciding how complex an MVP cannot be done formulaically
 - It requires judgment!
 - When in doubt, simplify
 - Avoid overbuilding and overpromising
 - Any additional work beyond what needs to get you starting the loop might be a waste
- An MVP is not just a prototype, its aim is to test the business hypotheses.
 - It serves in providing the needed dose of reality.
 - the goal of the MVP is to begin the process of learning.
- Design and quality can be a stumbling block for founders. Everyone wants to create something they can be proud of.
 - If you don't know who they customer is, you don't know what quality is.
 - If you are not ashamed of your version 1.0, you waited to long.

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Speed Bumps in Experimenting with an MVP



- Five most common challenges in developing an MVP include:
 1. legal issues (IP protection),
 2. Fears about competitors
 3. Branding Risk
 4. impact on morale.
 5. Audacity of zero
- 1. **Legal issues:**
 - Due to some established regulations, you might not be able to start experimenting!
 - **Example:** Lyft was about to be shutdown in California (it received several *cease and desist* in 2012 from several governments)!
 - Alongside, you may want to rely on *patent protection*, especially *if* a new scientific breakthrough is at the heart of your startup's competitive advantage
 - This is worth considering, albeit the delay that might arise
 - All of the above might truly delay your learning benefits

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Speed Bumps in Experimenting with an MVP



2. Fears about competitors:

- This is one of the most common objections for experimenting with an MVP (*someone will steal our idea, especially large companies!*)
 - **Test:** Call an established company and try to get it to steal your idea
 - The truth is that most of big companies are already flooded with good ideas
- Sooner or later, if you are successful, you will face competition from *fast followers* (time spent in *stealth mode* is unlikely to avoid this!)
- The only way to win is to learn faster than others, which requires experimenting!

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Speed Bumps in Experimenting with an MVP



3. Branding risks:

- If you plan to invest in building a great brand and experimenting with an MVP can seem like a dangerous risk, launch the MVP under a different brand name
- Do not engage in *vocal launch activities* such as PR and media hype during your initial MVP experimentations
 - Interestingly, as a startup, you have the advantage of being obscure (*unlike some big companies*)
- Engage with a public marketing launch only after your product has proved itself with real customers

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Speed Bumps in Experimenting with an MVP



4. **Impact on morale:**
 - MVPs often result in bad news!
 - This should not demoralize you whatsoever
 - In fact, this is a great news (or a *gift*) in reality
 - Use this gift to figure out the exact problems in your plan
 - Tackle these problems systematically, conduct related thoughtful experiments, and measure progress accordingly
 - Finally, *persevere* or *pivot* (more on this shortly)

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Audacity of Zero



5. **Audacity of Zero**
 - It is often easier to raise money when you have:
 - *zero revenue*
 - *zero customers*
 - *zero traction*
 - than when you have a small amount of each!
 - Zero invites imagination, but small numbers invite questions about whether large numbers will ever materialize
 - This phenomenon (called *the audacity of zero*) creates a brutal incentive:
 - Postpone getting any version of a product out until you are certain of success

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Audacity of Zero



- Audacity of zero
- If you postpone experimenting with your MVP, some unfortunate results will emerge like:
 - The amount of wasted work may increase
 - Essential feedback will be missed
 - The risk that your startup will build something nobody wants may increase
- But funding is important (*a dilemma!*), so what is the way out?
 - Tradeoffs:
 - Would you prefer to attract venture capital and potentially squander it?
 - Or, would you prefer to attract venture capital and wisely use it?
 - Use an MVP to *experiment* (initially, *silently*) with early adopters within your beachhead market
 - Verify your concept via testing ALL its elements, *starting with the riskiest ones*
 - Scale out on a *solid slate*— *funds will come naturally!*

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Experiment and Data capture



- At the beginning all a startup has is a just a business model.
 - Financial projections are usually an ideal and often far removed from the reality of the early days.
 - Very early on, the startup needs to do two things.
 - Measure current state (Where are we right now) and confront the herd truths that assessment reveals.
 - Devise experiments to learn how to move the real numbers closer to the ideal reflected on the business plans.
- Most products—even the ones that fail—do not have zero traction.
 - Most products have some customers, some growth, and some positive results.
 - One of the most dangerous outcomes for a startup is to bumble along in the land of the living dead.
- As a startup, how do you know if what you are doing (product development) is leading to the outcome you are seeing (Traction)?
 - Innovation Accounting

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Innovation accounting



- Innovation accounting enables startups to prove objectively that they are learning how to grow a sustainable business.
 - (Revenue and Profits are usually not good indicators of progress at this stage)
 - Identify and quantify critical assumptions, show your progress towards confirming your assumptions.
- Innovation accounting works in three steps:
 - use a minimum viable product to establish real data on where the company is right now (**baseline**)
 - Improvement of MVP (fine tune product/service) towards the ideal. Test and collect data and keep making changes based on feedback.
 - Pivot or persevere. Look at your progress, decide if to pivot (product pivot/customer/revenue model) or persevere which means to keep refining your product and processes until you grow your market.
- The process of gathering and analyzing the data is a primary market research approach.

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PIVOT (OR PERSEVERE)



- Every entrepreneur eventually faces an overriding challenge in developing a successful product: deciding when to pivot and when to persevere.
- All the data collected with the MVP should help a startup understand if its making sufficient progress to believe its original hypotheses is correct or if there is need to make a major change, i.e.. Pivot.
- Collect good data!
- Know your runway.
 - the amount of time remaining in which a startup must either achieve lift-off or fail.
 - When startups start to run low on cash, they can extend the runway two ways: by cutting costs or by raising additional funds.
- Pivots require courage
 - most entrepreneurs who have decided to pivot and they will tell you that they wish they had made the decision sooner.

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PIVOT (OR PERSEVERE)



- Pivots require courage
- Three reasons why entrepreneurs delay in pivoting.
 - Vanity metrics. You live in a bubble until you can't.
 - You don't know what hypothesis to test.
 - You are just scared. Acknowledging failure can lead to dangerously low morale.

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Leap-of-Faith Assumptions



The riskiest elements of a startup's plan/concept (i.e., the parts on which everything depends) are called *leap-of-faith assumptions*

E.g., What was the main leap-of-faith assumption of Dropbox?

- File synchronization is a problem

• **Note:** Most people do not know about a certain solution (or even a problem); but once they experience the solution, they cannot imagine how they ever lived without it!

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Leap-of-Faith Assumptions



The two most important leap-of-faith assumptions of any startup are the value hypothesis and the growth hypothesis

The value hypothesis:

- It tests whether the product is really delivering value to customers after they start using it
- A testing metric: retention rate

The growth hypothesis:

- It tests how new customers will discover the product
- A testing metric: referral rate or Net Promoter Score (NPS)

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Example: Facebook



In 2004, Facebook had 150,000 registered users with very little revenue

Yet, that summer they raised their first \$500,000 in venture capital

Less than a year later, they raised an additional \$12.7 million

How Facebook was able to raise so much money when its actual usage was small?

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Example: Facebook



- To answer this question, it suffices to look at Facebook's value and growth hypotheses:
 - *Validated value hypothesis:*
 - More than half of the users came back to the site every single day
 - *Validated growth hypothesis:*
 - Facebook launched on Feb 4, 2004, and by the end of that month, almost $\frac{3}{4}$ of Harvard's undergraduates were using it (*without spending a dollar on marketing or advertising!*)

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